

2.2 Apply Structured Prompt Templates

Scenario

In this lab you will learn how structured prompt templates can standardize AI outputs, reduce ambiguity, and enforce professional formats in security operations. By working with firewall logs, you will see how predefined templates guide analysis and reporting, making it easier to compare results across analysts. You will also practice completing partially filled templates, correcting flawed examples, and adapting templates for your own needs.

Your Mission:

- Apply structured prompt templates to real security log data.
- Compare freeform prompts against structured templates.
- Fill in and customize template fields to enforce consistency.

Exam Objectives

This activity is designed to test your understanding of and ability to apply content examples in the following CompTIA SecAI+ objectives:

- 1.1 Compare and contrast various AI types and techniques within the context of cybersecurity.
- 2.2 Given a set of requirements, implement security controls for AI systems.

2.2.1 Security Alert Summary

A common SOC workflow is documenting suspicious activity in a standardized format. The Security Alert Summary Template ensures no critical detail is missed and every analyst reports in the same style.

Step 1: Open **Firefox** and sign into **Open WebUI**.

Step 2: From the **New Chat** page, select the **model** dropdown menu and select the **mistral:7b** model.

Step 3: In the chat, enter the following prompt and note the results.

Prompt 1:

You are a SOC analyst. Read the INCIDENT CARD and write a concise paragraph describing what happened in plain language. Do not add facts not present in the card.

INCIDENT CARD (PHISHING - FINANCE)

- Date (CT): 2025-09-14
- First observed: 09:12:33
- Last observed: 10:05:11
- Targets: Finance staff (~25 inboxes)
- User reports: 3 (two clicked, one entered credentials is unconfirmed)
- Subject (examples): "Payment Advice - URGENT", "Invoice Past Due"
- From: accounts@billingsafe.co (Return-Path: bounce@billingsafe.co)
- Domains: billingsafe.co (registered 2025-09-10)
- Auth results: DMARC=fail, SPF=softfail, DKIM=none

- Links: `https://invoice-billingsafe[.]co/login` (Microsoft 365 lookalike)
- Attachments: `PaymentAdvice.htm` (HTML)
- Controls: Initial quarantine applied; gateway rule updated; some emails reached users
- Known impact: None confirmed; credential theft possible but unverified

Insights: Asking an open-ended question gives you a baseline. You see what the model produces without guidance-its natural level of detail, structure, and terminology. This makes it easier to appreciate the improvements once you apply a template, because you'll have something unstructured to compare against.

Step 4: In the same chat, enter the following prompt and note the response.

Prompt 2:

Complete the SECURITY ALERT SUMMARY TEMPLATE strictly from the INCIDENT CARD. If a field is unknown, write UNKNOWN.

SECURITY ALERT SUMMARY TEMPLATE

THREAT CLASSIFICATION:

- Severity Level: [CRITICAL/HIGH/MEDIUM/LOW]
- Threat Type: Phishing (Credential Harvesting)
- Confidence Level: [HIGH/MEDIUM/LOW]

TECHNICAL SUMMARY:

- Source: [sender/return-path]
- Target: [teams/departments]

- Attack Vector: Email, Link to fake login
- Affected Services: Email, Authentication

INDICATORS OF COMPROMISE (IOCs):

- Sender Domain(s): [list]
- URL(s): [list]
- Auth Results: [DMARC/SPF/DKIM]
- Attachment(s): [name/type]

TIMELINE:

- First Observed: [time]
- Last Observed: [time]
- Duration: [compute or UNKNOWN]

IMPACT ASSESSMENT:

- Systems Affected: [range or UNKNOWN]
- Data at Risk: Credentials (POSSIBLE)
- Service Disruption: Minimal

RECOMMENDED ACTIONS:

- 1.
- 2.
- 3.

NEXT STEPS:

- 1.
- 2.
- 3.

Insights: This template is designed for SOC analysts and incident responders. It standardizes how security events are documented so that every alert includes key elements: classification, technical details, indicators of compromise, timeline, impact, and recommended actions. By enforcing this structure, the template ensures that alerts are complete, consistent, and actionable, regardless of which analyst prepares them.

You've successfully implemented a Security Alert Summary template.

2.2.2 Incident Triage Template

When multiple alerts come in, analysts need a fast way to sort the noise from the real threats. The Incident Triage template helps capture the essential details right away so incidents can be classified, prioritized, and escalated if necessary.

Step 1: In the same mistral chat, enter the following prompt and note the response.

Prompt 1:

Complete the INCIDENT TRIAGE ASSESSMENT strictly from the INCIDENT CARD. Use the rubric: P2-HIGH if multiple users affected and credential theft is plausible but unconfirmed.

INCIDENT TRIAGE ASSESSMENT

INCIDENT METADATA:

- Incident ID: INC-2025-0914-FIN-01
- Date/Time Reported: 2025-09-14T10:15:00 CT
- Reporter: SOC queue (user reports escalated)
- Initial Source: Email Security Alerts + User Reports

INCIDENT CLASSIFICATION:

- Category:
- Subcategory:
- Severity: [P1/P2/P3/P4]
- Urgency: [IMMEDIATE/4HR/24HR/48HR]

SCOPE ASSESSMENT:

- Affected Systems:
- Affected Users:
- Affected Services:
- Geographic Scope:

TECHNICAL INDICATORS:

- Attack Vectors:
- Suspicious Domains:
- URLs:
- Attachments:
- Auth Results:

BUSINESS IMPACT:

- Service Availability: Operational
- Data Confidentiality: POTENTIALLY_BREACHED
(credentials)
- System Integrity: Intact
- Regulatory Implications: UNKNOWN

Insights: This template is used by SOC analysts and shift leads during the first pass of an investigation. Its purpose is to quickly classify an incident, estimate scope, assign severity and urgency, and decide whether it needs to be escalated. By capturing this information in a consistent format, teams can prioritize incidents effectively and ensure the most critical threats are handled first.

You've successfully implemented an Incident Triage template.

2.2.3 Executive Summary Template

Technical details are vital for analysts, but executives need information framed in terms of business impact and decisions. The Executive Summary template condenses complex incidents into plain language, highlighting only the most critical findings, risks, and recommended actions.

Step 1: In the same mistral chat, enter the following prompt and note the response.

Prompt 1:

Produce a CYBERSECURITY EXECUTIVE SUMMARY for leadership, based strictly on the INCIDENT CARD.

CYBERSECURITY EXECUTIVE SUMMARY

EXECUTIVE OVERVIEW:

[2 sentences, plain language, no acronyms.]

KEY METRICS:

- Incidents Detected:
- Incidents Resolved:
- Mean Time to Detection:
- Mean Time to Response:
- Systems Affected:

BUSINESS IMPACT:

- Immediate:
- Potential:
- Financial:

ACTIONS TAKEN:

- Quarantined messages and tightened gateway rules
- Blocked sender domains/URLs
- User notifications initiated

RECOMMENDATIONS:

- 1.
- 2.
- 3.

Insights: This template is written for executives and senior decision-makers. It strips away technical jargon and focuses on business impact, risk, and next steps. By presenting incidents in a concise, plain-language format, it ensures leadership can make informed decisions quickly without needing to parse detailed logs or analyst notes.

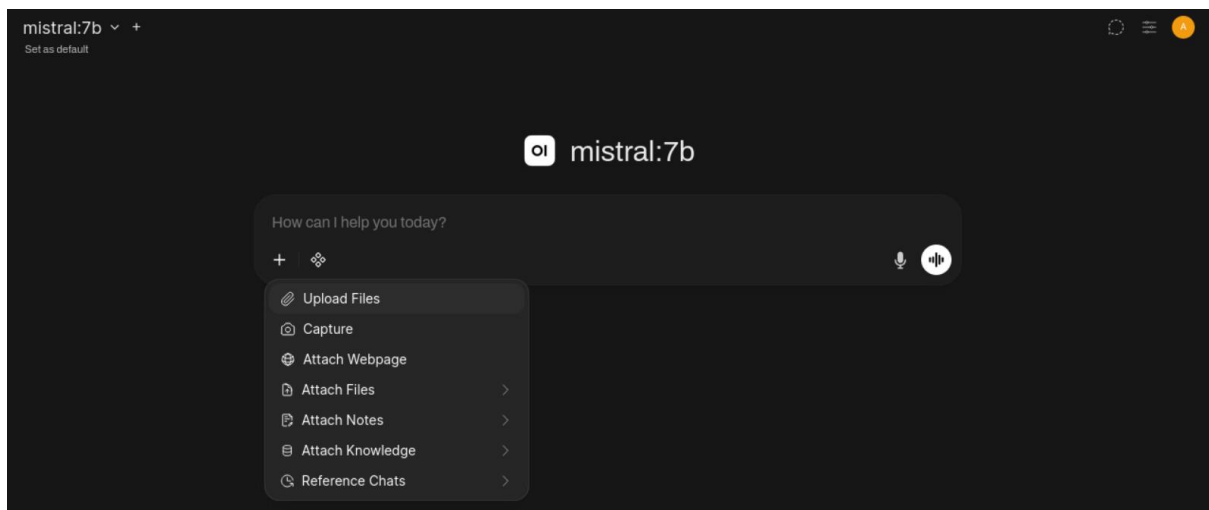
You've successfully implemented an Executive Summary template.

2.2.4 Prompt Templates with Internal Data

Now that you've practiced templates on general scenarios, it's time to apply one to real security data. In this step, you'll upload an alert file and use a structured prompt template to generate a standardized alert summary from actual network activity.

Step 1: In OpenWebUI, start a new chat with mistral.

Step 2: Beneath the chat box, select **+ More** and then select **Upload Files**.



Step 3: Select **Alert File1.json** from the **Labfiles** folder, then select **Open**.

Step 4: In the chat, enter the following prompt and note the response:

Prompt 1:

Analyze the json file, output:

- 1) Summary (2 sentences) using info, analysis, threat_level_id, tags.
- 2) Key IOCs (max 5 bullets).
- 3) Recommendations (3 short bullets).

If something is missing, write UNKNOWN. Do not add facts.

Insights: When you upload a large file straight into the model, it has to read the whole dataset in one go. This makes responses slow, can cause the model to skip or oversimplify details, and often leads to inconsistent outputs. In practice, security teams use Retrieval-Augmented Generation (RAG): files are broken into indexed chunks, and the model only pulls in the most relevant pieces. RAG makes analysis faster, more accurate, and scalable, which you'll explore in a later lab.